# Intelligent Customer Churn Prediction Using Ensemble Learning and Explainable AI

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***Abstract*—Churn prediction is very important for subscription-based businesses because keeping the existing customers is more cost-effective than attracting new ones. In this study, the intelligent customer churn prediction system is proposed using explainable artificial intelligence and ensemble machine learning algorithms. Several classifiers have been trained and tested using a telecommunications company data set that contained attributes like financial, behavioral, and demographic factors. The metrics used in evaluation included precision, recall, F1-score, and ROC-AUC. The XGBoost model is seen to be performing better in determining the risk of churning consumers, scoring higher in recall, which is important to prevent customer loss. Moreover, the SHAP method was applied to analyze the outputs of the model and find influential attributes for churn prediction. Cost-sensitive threshold tuning was introduced in order to make the model more aligned with the business objectives and optimize recall scores. It can be concluded that price, customer satisfaction, and service engagement play a vital role in predicting turnover.**

***Index Terms*—Customer Churn, Machine Learning, Ensemble Learning, XGBoost, Explainable AI, SHAP, Cost-Sensitive Learning**

## I. Introduction

Retaining customers has been a big challenge in competitive sectors such as telecommunications because of its significance in profitability. "Customer churn," which entails cessation in the utilization of services and, as such, represents a loss of revenue for the company, has emerged as a serious problem in contemporary organizations. Proactive actions in keeping clients are possible in case of expected turnovers.

The growing generation of customer data has made machine learning techniques an efficient way for predicting churn. Classification performance is usually regarded as a major focus in traditional approaches. However, many times, they do not address practical concerns such as the recognition of high-risk clients and understanding the reasons for customer attrition.

The principal aim of this study is developing a predictive system for churn using explainable AI to provide interpretation along with advanced ensemble learning approaches to predict customer attrition. Moreover, cost-sensitive optimization is considered to ensure that the developed model aligns with real-life business objectives, hence ensuring effective identification of risky clients.

## II. Literature Review

Machine learning-based methods has been widely adopted in churning prediction in multiple research and business actions. An important point made regarding the integration of business goals in the process was that of Verbraken et al., (2012) [1] which introduced the concept of Expected Maximum Profit (EMP), emphasizing that the performance of a model should be evaluated in terms of profit and not just accuracy. With regard to tackling the problem of data imbalance, Burez & Van den Poel (2009) [3] recommended the use of cost sensitive evaluation metrics and AUC.

As far as feature engineering goes, it can be argued that this is an important part of improving the accuracy in predicting churn. As per Huang et al., (2012) [4], different kinds of features from customer behavior, billing information, and services utilization can help boost the accuracy of prediction. An interesting point discussed in relation to prediction was that of data uncertainty by Amin et al., (2019) [2]. There would be no uniformity in prediction certainty for all types of data.

The new trends focus on the concept of developing hybrid and ensemble methods for predicting churns. In their study, Joy et al. (2024) [5] created a hybrid method involving deep learning and boosting methods resulting in the accurate outcomes. It was proven in the research that it is important to apply the ensemble algorithms including Random Forest, Gradient Boosting, and XGBoost for building models for predicting churn.

It is important to implement XAI (Explainable Artificial Intelligence) in order to improve predictability. El Attar and El-Hajj (2026) [6] and Asif et al. (2025) [7] constructed ensemble models using SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) methods to detect the important factors affecting the churn including tenure, contracts, and engagement. Raut and Puri (2026) [8] proved once more the importance of applying XAI

in their study using SHAP-based method in their predictive modeling.

Despite all of these accomplishments, research tends to focus solely on one aspect of prediction or explainability without incorporating any optimization for business purposes. The absence of this crucial component has been highlighted by Sutar et al. (2025) [9] and Manzoor et al. (2024) [10] in their review articles.

To address this issue, the current study aims to fill this research gap by developing an intelligent churn prediction model with ensemble methods (Random Forest, Gradient Boosting, and XGBoost), as well as SHAP for explainability and threshold optimization based on costs. Thus, this approach allows achieving high accurate prediction while at the same time being consistent with business objectives.

## III. Methodology

The methodologies used include data preprocessing, modeling, evaluation, and model interpretability techniques.

The initial data set had a total of 50 columns. Several columns were removed from the data before training since the columns were unique identifier columns, geographic columns, and post churn leakage columns. Some of the columns removed include Customer ID, Country, State, City, Zip Code, Latitude, Longitude, Customer Status, Churn Score, Churn Category, Churn Reason, and CLTV. Only two columns, Offer and Internet Type, had missing values in the dataset, and they were imputed using "No Offer" and "None" for Offer and Internet Type, respectively. Churn Label was mapped using 1 and 0 for Yes and No, respectively. Finally, one hot encoding of all the other columns, which were categorical columns, was done using get_dummies with drop_first as True, giving 45 features, and the entire dataset was made into an integer datatype.

After splitting the dataset into training and testing sets based on an 80-20 ratio, three different ensemble-based models including Random Forest, Gradient Boosting, and XGBoost were used to predict the customer churn problem. The models were chosen for their capability to detect non-linear relationships.

A cost-sensitive approach was adopted through manipulation of the classification threshold to incorporate business-oriented decision making. In an attempt to improve the ability to predict which customers are likely to leave, the default threshold value of 0.5 was reduced to 0.3. This approach values recall higher as it is costly to not predict the churners.

Finally, the results from the model were analyzed to determine the importance of each predictor in predicting the likelihood of churn.

## IV. Experimental Setup

The tests were performed using Python within the Google Colab workspace. Libraries such as scikit-learn and XGBoost were used in model development, while Pandas and NumPy were applied in data preprocessing. In addition, SHAP was used for interpretability analysis.

The dataset containing 7043 customer records includes features regarding demographics, usage, billing, customer engagement, and others in both numerical and categorical form. The data preprocessing process was implemented after which it was transformed into all-numerical form.

80% of the data was used in the training of models, while the remaining 20% was used for testing purposes. Several metrics were selected in order to evaluate the performance of models, including precision, recall, F1 score, and ROC-AUC.

## V. Results and Discussion

The performance evaluation of the classification models was done using precision, recall, F1-score, and ROC-AUC metrics. Table I displays the summarized results.

TABLE I
Performance Comparison of Classification Models

| Model | Precision (Churn) | Recall (Churn) | F1-Score | ROC-AUC |
|---|---|---|---|---|
| Random Forest | 0.98 | 0.84 | 0.90 | 0.979 |
| Gradient Boosting | 0.95 | 0.86 | 0.91 | 0.988 |
| XGBoost | 0.94 | 0.88 | 0.91 | 0.986 |

A total of 1409 samples comprised the test dataset, where 1009 were non-churns and 400 were churns, thus representing a ratio of roughly 72% to 28%. All models yielded a similar overall accuracy level of around 95 percent. Overall accuracy cannot be used solely when assessing churn predictions owing to class imbalance. One of the main differences between all the models lay in the recall rate of the churns. Compared to the other models, Random Forest yielded the highest precision of 0.98 for the churn class, although it had the lowest recall rate of 0.84, implying that it did not capture the majority of churns. Recall rate was increased to 0.86 by Gradient Boosting with a slight reduction in precision to 0.95. The XGBoost produced the highest recall rate of 0.88 at a default threshold value of 0.5 and was therefore the best suited for minimizing the number of missed churns. Churn recall was further improved to 0.90 after tuning the threshold of XGBoost from 0.5 to 0.3.

Table I represents the aggregated accuracy of all three ensemble techniques used in the work. In regard to the task of predicting churn behavior, the algorithm of random forest turned out to have extremely high precision, while its recall turned out to be low, meaning that this model cannot recognize those customers that are likely to churn. Based on the highest ROC-AUC metric, gradient boosting became a technique that was the best at distinguishing churn vs. non-churn customers. But, at the same time, XGBoost managed to detect more churners than other algorithms due to having the highest recall value in this case.

XGBoost is selected as the best algorithm because the task requires minimizing the number of false negatives. The model was additionally aligned with business requirements using cost-sensitive threshold tuning.

As can be observed in Table II, an increase in the value of the recall metric is observed, making it easy to identify

TABLE II
EFFECT OF THRESHOLD TUNING ON MODEL PERFORMANCE

| Threshold | Precision | Recall | F1-Score |
|---|---|---|---|
| 0.5 | 0.94 | 0.88 | 0.91 |
| 0.3 | 0.92 | 0.90 | 0.91 |

those customers who would abandon the company's services at the decreased threshold from 0.5 to 0.3. Additionally, there is a small decrement in the precision metric, but for churn detection purposes, it is a good exchange.

To further analyze the performance of the models, confusion matrices were used for all three models. All the models performed well when it came to classifying the cases into churn and non-churn customer segments.

However, on further investigation, it was seen that Random Forest performed exceptionally well in the case of correctly predicting non-churn cases, whereas Gradient Boosting and XGBoost managed to balance both the churn and non-churn cases efficiently. On comparing the models, it was observed that XGBoost performed comparatively better in the case of predicting churn customers.

It should be noted that the focus should be on minimizing false negative cases while predicting churn customers since they may lead to the loss of potential customers. Hence, among the models, XGBoost had lesser numbers of false negatives than other models.

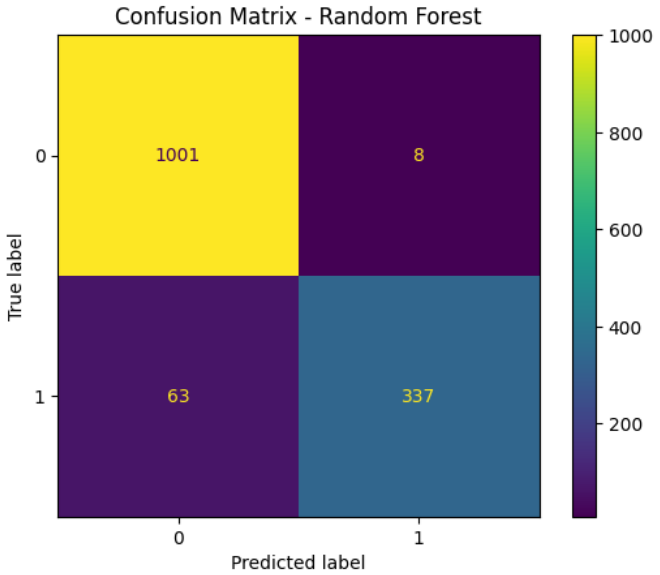


Fig. 1. Confusion Matrix for Random Forest Model

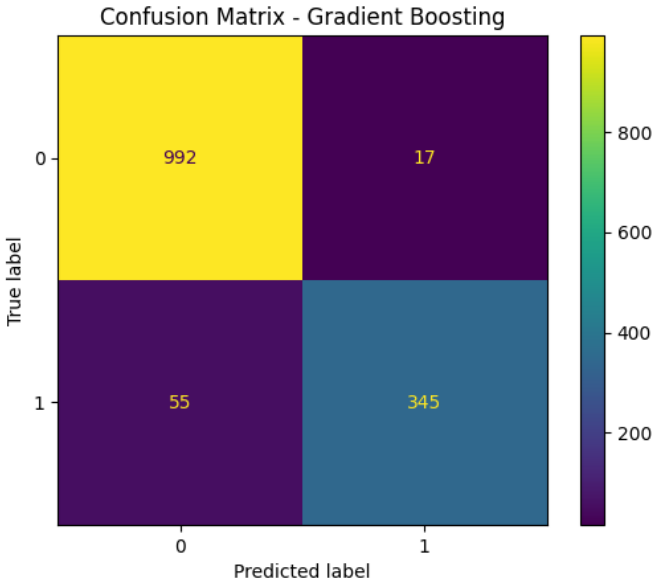


Fig. 2. Confusion Matrix for Gradient Boosting Model

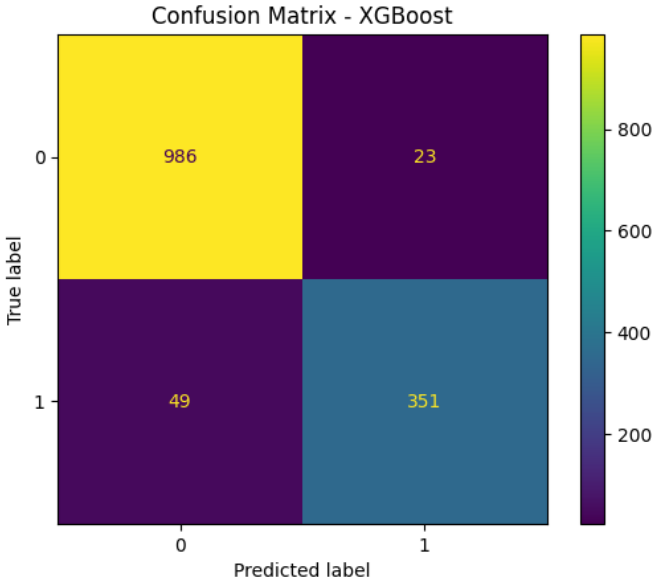


Fig. 3. Confusion Matrix for XGBoost Model

ROC curves were utilized in analyzing the discrimination power of all models. All models showed high AUCs greater than 0.97. The model with the highest value is Gradient Boosting at 0.988, while XGBoost and Random Forest had scores of 0.986 and 0.979 respectively.

While Gradient Boosting performed slightly better in terms of separation between both classes, there are very slight differences among the three. Therefore, all ensemble models can be said to successfully address the issue of predicting customer churn.

Although the ROC-AUC gives an average evaluation of the model, it does not indicate any trade-off for each category.

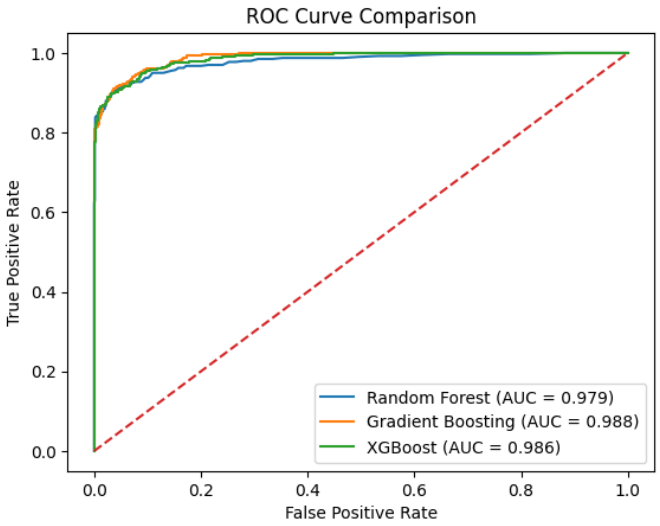


Fig. 4. ROC Curve Comparison of Models

From the confusion matrix, it can be seen that the XGBoost has done better by minimizing false negatives, which aligns more with the aim of the problem of predicting churn. Hence, even though Gradient Boosting performs slightly better in terms of AUC scores, XGBoost is a better choice.

This is yet another confirmation that choosing XGBoost is the right way to go since it not only performs well in general but also manages to pick out clients prone to churn.

The interpretation of the predictions made by the models included the use of explainable AI techniques. In order to find the most important features contributing to customer churn, a SHAP summary plot was made.

As can be observed in Fig. 5, one of the critical aspects of retention is the level of satisfaction of the client. It is evidenced by the high degree of satisfaction greatly reducing the chances of churn and vice versa.

Furthermore, another critical effect is the number of referrals since the higher it is, the greater the loyalty of the client is. On the contrary, the lower the number of referrals, the higher the chances of churn. The absence of additional services, such as online security, has a positive effect on churn since clients that do not use value-added services tend to churn.

Customer attrition is positively related to financial matters, such as monthly payments. Moreover, customers with two-year contracts have lower chances of attrition.

Other important effects are the number of dependents and utilization features, including data consumption and subscriptions. However, they play a much lesser role in customer attrition than the previous ones.

As a result of applying SHAP, important insights have been identified as regards consumer behavior, which have proved to be the most significant factors affecting customer attrition; these factors include price, engagement, and satisfaction. The approach under discussion is applicable to real-world conditions, as it combines explainable artificial intelligence with cost-sensitivity.

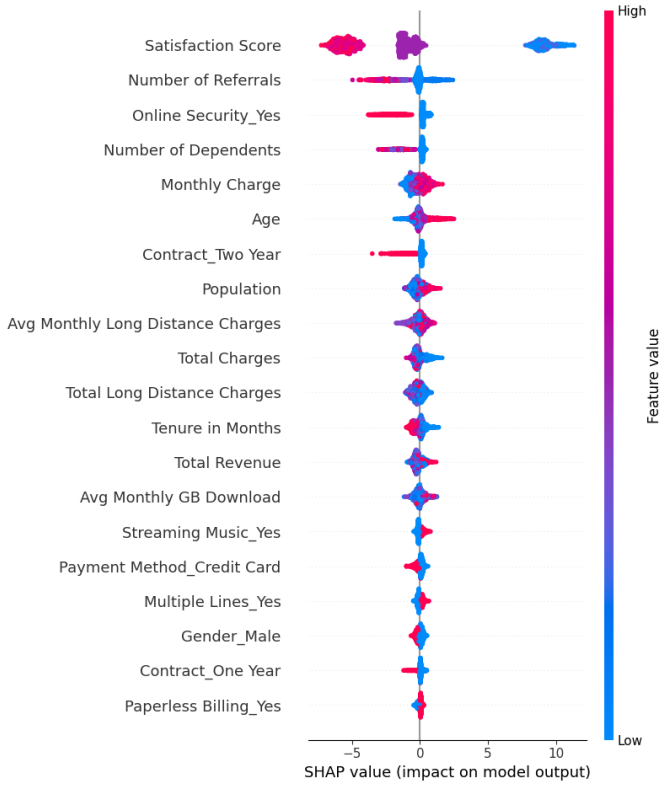


Fig. 5. SHAP Summary Plot Showing Feature Importance and Impact on Customer Churn

## VI. Conclusion

In this paper, an intelligent customer churn prediction system that incorporates explainable artificial intelligence, ensemble machine learning methods, and cost sensitivity optimization was proposed. With regards to the task at hand, the proposed method was successful in addressing issues of forecast accuracy and practical business considerations.

As a result of outstanding performance and high recall rate, XGBoost was selected as the top-performing model. The performance in terms of minimization of false negatives was also enhanced by the model. This is because in the case of a churn prediction problem, missing out on a potential churner results in losses for the organization. Though Gradient Boosting scored marginally higher in terms of ROC-AUC, XGBoost had more to offer than that. The ability of the algorithm to accurately pinpoint customers who will churn makes it particularly valuable in minimizing potential revenue loss. The focus on recall and the application of cost sensitivity optimization helped improve the effectiveness of the model and make it align with practical business considerations.

The application of explainable AI through SHAP proved to give greater insight into the components that play a role in determining customer attrition. From the study, it was evident that some of the major components affecting customer attrition were happiness, price, utilization of services, and

customer engagement. This information can be applied to create specific customer retention strategies, such as enhancing client engagement, customized pricing, and improving service quality.

The proposed approach proves the effectiveness of integrating advanced machine learning techniques with interpretability and optimization towards a business perspective. From the research, it is evident that such techniques can be applied to develop reliable decision support systems for client retention.

To enhance the model's prediction accuracy, in future studies, it will be essential to incorporate time-dependent behaviors and live streaming data. The practicality of the system can also be improved by integrating it with customer relationship management (CRM) systems and organizational dashboards.

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